Implementation of Convolutional Neural Network for ECG-Based Arrhythmia Classification

Tugas Akhir

diajukan untuk memenuhi salah satu syarat

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LEMBAR PENGESAHAN

Implementasi Convolutional Neural Network untuk Klasifikasi Aritmia Berbasis EKG

Implementation of Convolutional Neural Network for ECG-Based Arrhythmia Classification

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Tugas akhir ini telah diterima dan disahkan untuk memenuhi sebagian syarat memperoleh gelar pada Program Studi Sarjana S1 Informatika

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LEMBAR PERNYATAAN

Dengan ini saya, M Luthfi Aditya Gunandi, menyatakan sesungguhnya bahwa Tugas Akhir saya dengan judul **Implementation of Convolutional Neural Network for ECG-Based Arrhythmia Classification** beserta dengan seluruh isinya adalah merupakan hasil karya sendiri, dan saya tidak melakukan penjiplakan yang tidak sesuai dengan etika keilmuan yang belaku dalam masyarakat keilmuan. Saya siap menanggung resiko/sanksi yang diberikan jika di kemudian hari ditemukan pelanggaran terhadap etika keilmuan dalam buku TA atau jika ada klaim dari pihak lain terhadap keaslian karya,

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Yang Menyatakan

M Luthfi Aditya Gunandi

Implementation of Convolutional Neural Network for ECG-Based Arrhythmia Classification

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Abstract

Electrocardiogram (ECG) signals are vital for detecting arrhythmias, abnormal heart rhythms that can lead to serious health complications. Traditional ECG classification methods often rely on extensive feature engineering and expert input. This study used the MIT-BIH Arrhythmia dataset to assess the performance of Convolutional Neural Network for automated arrhythmia classification. To boost generalization, CNN's design includes a few layers of complexity, and dropout. To address class imbalance, data balancing techniques are applied. Time Warping and Magnitude Warping augmentation is introduced to simulate signal variability and further improve robustness. A comprehensive hyperparameter tuning framework evaluates five CNN configurations, with the optimal setup using three convolutional layers (128-256-128 filters), batch normalization, dropout (0.15–0.4), and two dense layers (256-128 units). Achieving an accuracy and F1-score of 97.41% from the baseline, highlighting its potential for real-world automated ECG analysis. This work extends deep learning approaches for biological signal processing, promoting more efficient and accurate arrhythmia detection.

***Keywords**: Convolutional Neural Network, ECG Classification, Arrhythmia Detection, Deep Learning, MIT-BIH*

1. Introduction

Heart failure constitutes one of the biggest reasons for funerals globally, with arrhythmia being one of the main types of heart disorders[1]. Arrhythmia is a condition in which the heart beats irregularly—too fast, too slow, or erratically—which can result in severe health complications if not detected early. Electrocardiogram (ECG), a non-invasive tool that analyzes rhythmic activity on a daily basis, is one of the most effective ways to identify arrhythmias[2]. Manual interpretation of ECG signals requires expertise and is prone to human error. With the advancement of artificial intelligence, particularly deep learning, automatic arrhythmia detection using ECG signals has gained significant attention. Convolutional Neural Networks (CNNs), a class of deep learning models, have shown superior performance in detecting patterns from raw ECG signals, eliminating the need for handcrafted features. This allows CNNs to be used effectively in the classification of various types of arrhythmias[3].

Convolutional Neural Networks (CNNs) have been extensively utilized in recent studies for arrhythmia classification due to their superior ability to automatically extract relevant features from ECG signals. For example, Ramkumar et al. proposed a multi-scale CNN model that significantly enhanced arrhythmia detection performance when evaluated on benchmark ECG datasets[4]. In addition, hybrid models combining CNNs with Long Short-Term Memory (LSTM) networks have been shown to effectively capture temporal dependencies within ECG sequences, thereby improving classification accuracy[5]. Previous studies related to arrhythmia classification using CNNs include research by Hapsari et al. (2023), which demonstrated that CNN models applied to ECG signals can achieve up to 99.79% accuracy, 99.71% sensitivity, and 98.35% specificity, indicating high reliability in arrhythmia classification[6]. Pratama and Abadi (2023) used CNNs to classify 17 types of arrhythmias from ECG images, achieving 81% accuracy, including classes such as normal sinus rhythm and pacemaker rhythm[7]. Another study by Nurriski and Alamsyah (2023) showed the effectiveness of Deep CNNs in detecting arrhythmias with 98.88% accuracy, also addressing the issue of class imbalance in the dataset[8]. Mohebbanaaz (2021) reported a CNN model reaching 98.70% accuracy[9].

This study employs a CNN-based algorithm to classify arrhythmia using ECG data. The classification includes multiple arrhythmia types, The model is assessed based on performance criteria such as accuracy, precision, recall, and F1-score. This objective is to improve the accuracy and effectiveness of arrhythmia detection and shows that CNN is an effective method for signal data.CNN. which can support medical practitioners in making faster and more accurate diagnoses. By automating ECG analysis through CNNs, This study intends to contribute to the development of smart healthcare solutions in cardiology. A confusion matrix is utilized to provide a comprehensive evaluation of the model's classification performance. It delivers detailed insight into the number of true positive, false positive, true negative, and false negative predictions for each arrhythmia class, enabling the identification of potential misclassifications and areas for model enhancement. This method of evaluation is particularly critical in medical diagnostics, where classes often carry unequal clinical importance, and incorrect classifications can lead to serious implications for patient care using [10].

2. Research Methods

2.1. Research Stages

To illustrate the research flow, a flowchart is needed. The flowchart consists of several steps to complete the research in order to achieve efficient results. The steps are loading the dataset, data preprocessing, data augmentation, reshaping data for CNN input, model creation and performance evaluation utilizing a Convolutional Neural Network (CNN).

The preprocessing stage consists of converting class labels, balancing class distribution, and one-hot encoding. In the data augmentation stage, it includes resampling minority classes and adding Time Warping and Magnitude Warping to ECG signals. In the evaluation stage, it includes plotting accuracy and loss, generating the confusion matrix, calculating ROC and AUC, and providing a classification report. as showed in Figure 1.

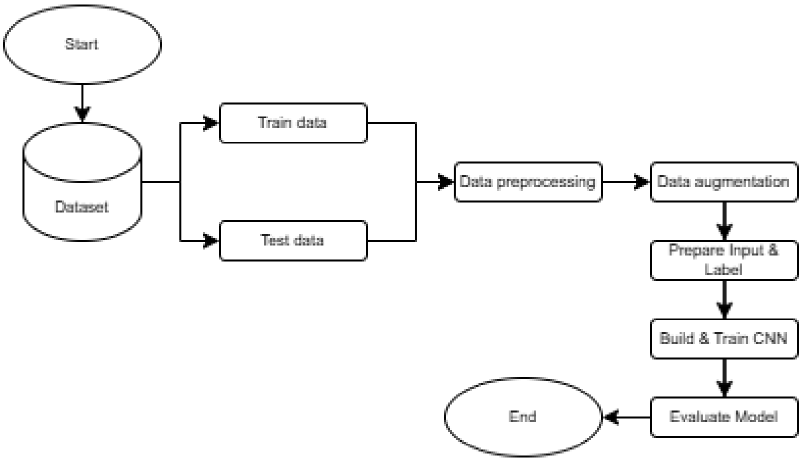

Figure 1. Flowchart system of this research.

2.2. Dataset

The MIT-BIH Arrhythmia Database comprises 48 thirty-minute segments of dual-channel ambulatory ECG recordings, collected from 47 individuals examined by the BIH Arrhythmia Laboratory between 1975 and 1979. Twenty-three recordings were randomly selected from a collection of 4000 24-hour ambulatory ECG recordings at Boston's Beth Israel Hospital; the additional 25 recordings were deliberately chosen from the same set to encompass less prevalent yet clinically significant arrhythmias that would be inadequately represented in a small random sample[2].

This dataset contains five heart rate classes (N, S, V, F, and Q) established by the AAMI standard[11] and classified using the MIT-BIH Arrhythmia Database, as showed in Table 1.

Table 1. The five main heartbeat classes.

Class	Total Test Data	Total Train Data
N	18118	72471
S	556	2223
V	1448	5788
F	162	641
Q	1608	6431

a. N (Normal Beat Group) shows the normal beat, LBBB, RBBB, atrial escape beat, and nodal escape beat.
b. S (Supraventricular ectopic pulse group) denotes atrial premature beats and abnormal supraventricular beats.
c. V (Ventricular ectopic pulse group) stands for premature ventricular contractions (PVCs) and ventricular escape beats.
d. F (Fusion Pulse Group): A mix of ventricular and regular beats.
e. Q (Unknown Pulse Group): Rhythms that do not belong in the preceding categories or cannot be categorize.

2.3. Preprocessing Data

The data preprocessing stage consists of several steps to prepare the dataset for training the CNN model. The steps are as follows:

a. Label conversion, the class labels in the dataset (located in the 188th column) are converted to integer type to ensure compatibility with one-hot encoding and model training.
b. Class distribution visualization, the original distribution of each class is visualized using pie charts to identify imbalances across arrhythmia classes. This visualization highlights the dominance of normal beats (class 0) over other classes, as showed in Figure 2.
c. One-Hot encoding, after balancing, the class labels are transformed into categorical format using one-hot encoding. This step is essential for classifying multiple classes with categorical a cross-en loss.
d. Data reshaping for CNN input, the ECG data, initially in 2D format, is reshaped into a 3D format of shape to meet the input requirements of 1D CNN layers.

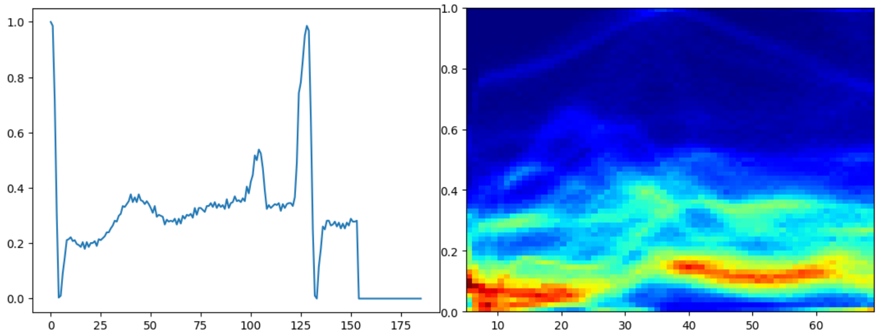

Figure 2. The visualization of distribution class on a single ECG data.

2.4. Augmentation Data

In the augmentation stage, the following procedures are applied to enrich the training set and correct class imbalance:

a. Due of the imbalance in class distribution, particularly for minority classes (Class 1-4), data balancing was achieved using the resample() technique, which involved upsampling or downsampling each training class to 20,000 samples and each test class to 10,000. This strategy contributes to fair representation of all classes, minimizing bias toward the dominant class and enhancing the model's capacity to identify trends in underrepresented classes. Random oversampling and undersampling are useful techniques for managing class imbalance and can promote fairness in model training. Recalibration is advised to maintain accurate probability estimations[12], as showed in Figure 3.

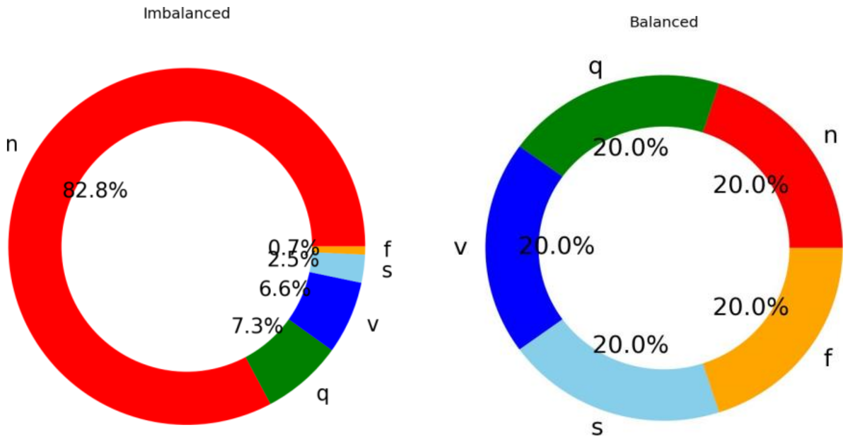

Figure 3. Percentage class of dataset before balancing (left) and after balancing (right).

b. Figure 4 depicts four ECG signal shapes: the original signal, the outcome of augmentation using the time warping approach, magnitude warping, and a combination of the two. Time warping alters the rate of change of the signal on the time axis, causing some patterns to look compressed or stretched, whereas magnitude warping alters the signal's amplitude to simulate intensity fluctuations. The combination of the two results in more complicated changes while preserving the signal's core properties. The below image compares each augmentation result to the original signal: time warping shows shifts in signal shape across specific time segments, magnitude warping shows clear differences in amplitude scale, and combined augmentation retains nearly identical shapes with subtle modifications. Thus, these techniques can be used to enhance training data diversity without losing the distinctive characteristics of the ECG signal.

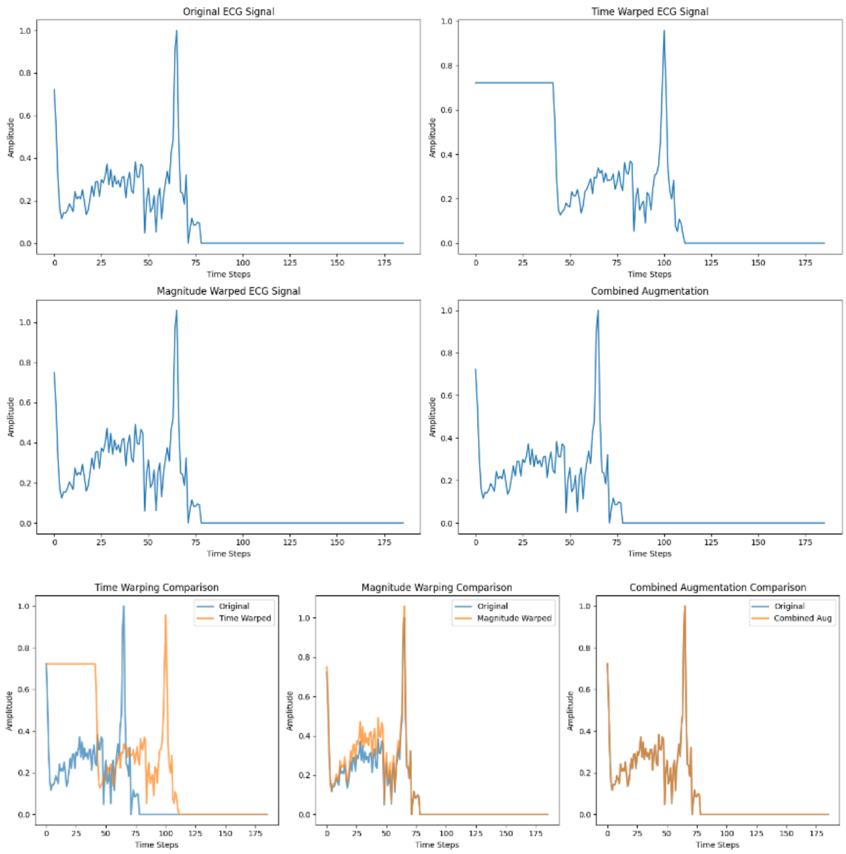

Figure 4. Visualization of a signal Augmentation and Comparison.

Based on the comparison in Figure 4, the time warping technique produces significant changes in the ECG signal time axis, causing certain amplitude patterns to shift and the waveform to undergo noticeable temporal distortion, especially at the beginning and peak of the signal. In contrast, magnitude warping preserves the original time position but modifies the amplitude scale variably across certain signal segments, resulting in waveforms that remain temporally consistent but exhibit intensity differences. These differences indicate that time warping primarily affects the speed or temporal distribution of the signal, while magnitude warping focuses on signal strength variations without altering the temporal structure. Combining the effects of time warping and magnitude warping simultaneously, so that the signal undergoes changes in both time distribution and amplitude scale.

2.5. Model Training

After the preprocessing and augmentation stages, the dataset was classified using a Convolutional Neural Network (CNN) architecture with adjustable hyperparameter configurations. CNN was chosen because it is capable of automatically learning spatial and temporal patterns in raw signal data such as ECG, without requiring manual feature extraction[10]. The local nature of convolution filters allows the model to capture important morphological characteristics in ECG signals, such as QRS complexes, P waves, and T waves, which are key indicators in arrhythmia classification. A normalization layer (batch normalization) is used to stabilize and accelerate the training process, while max pooling serves to reduce feature dimensions and prevent overfitting. The resulting feature map is then flattened and passed to a fully connected layer (dense layer), where dropout and L2 regularization are applied to improve the model's generalization ability. The output layer uses the softmax activation function to generate a probability distribution across five arrhythmia classes. The training process is performed using various optimizers (Adam, RMSprop, SGD) with a categorical cross-entropy loss function, and early stopping and model checkpointing are applied to prevent overfitting and preserve the best model weights.

Three main configurations were evaluated. The baseline model consists of three convolutional layers with 64 filters and kernel sizes of 6, 3, and 3, followed by max pooling, batch normalization, and two dense layers with 64 and 32 units, trained using Adam with a learning rate of 0.001. The optimized model increases the number of filters to 128–256, applies L2 regularization and a higher dropout rate, expands the dense layers to 256 and 128 units, and reduces the learning rate to 0.0008 for better training stability. The lightweight model is designed for computational efficiency, using only two convolutional layers (32 and 64 filters) with larger kernels, fewer dense units, no batch normalization, and trained using RMSprop with a learning rate of 0.002. Experimental results show a trade-off between accuracy, training time, and computational load, with the optimized model achieving the highest test accuracy.

2.6. Evaluation

This study evaluated the model's effectiveness in classifying ECG-based arrhythmias using a supervised learning approach. To ensure generalization capability, the dataset was partitioned into training and testing subsets using a train/test split method. A Convolutional Neural Network (CNN) was employed due to its capacity to automatically learn hierarchical features directly from raw ECG time-series signals, making it especially suitable for biomedical signal classification tasks[10].

The training performance was monitored using accuracy and loss metrics across epochs on both training and validation data. These learning curves help detect overfitting or underfitting tendencies during the training phase. To evaluate, forecasts on the test set are compared with the actual truth labels using a confusion matrix as given in Table 2.

Table 2. Standard form of a confusion matrix

Predicted/Actual	Predicted Postive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

A confusion matrix (Table 2) is a tabular representation that compares the actual target values with the model's predictions. It consists of four components:

a. True Positive (TP): A positive class that was accurately predicted.
b. False Positive (FP): A number that is projected as positive but is actually negative.
c. True Negative (TN): The prediction for the negative class was accurate.
d. False Negative (FN): A result that is actually positive but was mistakenly predicted to be negative

These numbers are important in computing performance measures like accuracy, precision, recall, and F1 score, which give a more detailed knowledge of the classifier's dependability, especially in unbalanced datasets[13]. Here are the definition of each performance metrics:

a. Accuracy is the model's overall accuracy.
b. Precision is the fraction of true positive predictions among all positive forecasts.
c. Recall (sensitivity) is the fraction of right positives among all true positives.
d. The F1-score is the harmonic mean of accuracy and recall.

The formula of each metrics can be found in several references, for instances in[14]. To further assess the classification capability. For every class, Area Under the Curve (AUC) values and Receiver Operating

Characteristic (ROC) curves are produced. The ROC curve illustrates the trade-off between true positive and false positive rates, while AUC quantifies this trade-off as a single value, where improved classification ability is indicated by a higher AUC. A higher ROC AUC indicates better performance. A perfect model would have an AUC of 1, while a random model would have an AUC of 0.5[15].

This multi-metric assessment provides a full understanding of CNN models' performance, notably their capacity to distinguish between distinct arrhythmia kinds, which is essential in medical diagnostics[16].

3. Result and Discussion

The Convolutional Neural Network (CNN) model developed to perform multi-class classification of electrocardiogram (ECG) data is thoroughly evaluated in this section. The primary objective of this study is to determine the CNN's ability to classify ECG data into five heartbeat categories: unknown (Q), fusion (F), ventricular ectopic (V), normal (N), and supraventricular ectopic (S)[2]. This study was conducted repeatedly with three different main configurations, namely Baseline, Optimized, and Lightweight, resulting in the Baseline experiment with the highest accuracy. This includes various general classification criteria, including accuracy, precision, recall, F1 score, confusion matrix, and multi-class ROC curve. These metrics provide a comprehensive overview of the model's overall performance and performance per class.

3.1. Model Comparison Results

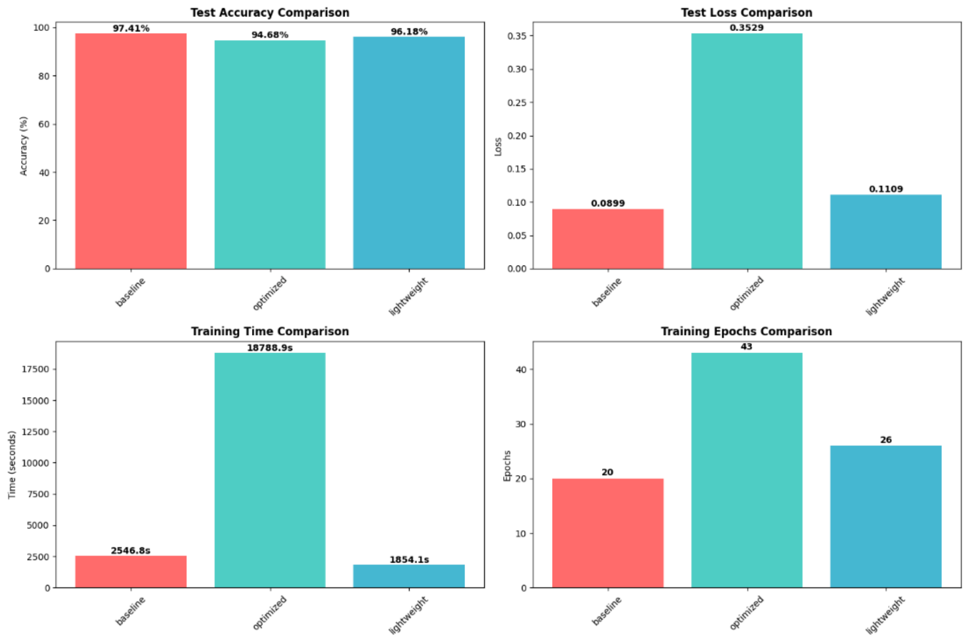

Figure 5. Model Comparison Results for Accuracy, Loss, Time, Epochs.

3.1.1. Test Accuracy Comparison

The Baseline model showed the highest accuracy of 97.41%, followed by the Lightweight model with 96.18%, and the Optimized model with 94.68%. This indicates that even though the Optimized model experienced an increase in the number of epochs, its accuracy was slightly lower than the other two models.

3.1.2. Test Loss Comparison

The lowest loss value was achieved by Baseline (0.0899), followed by Lightweight (0.1109), while Optimized had the highest loss (0.3529). The lower the loss, the better the model's ability to consistently predict test data.

3.1.3. Training Time Comparison

The Optimized model required the longest training time, 18,788.9 seconds, which is much longer than the Baseline (2,546.8 seconds) and Lightweight (1,854.1 seconds) models. This may be due to the complexity of the optimization process or parameter settings that caused the training process to take longer.

3.1.4. Training Epochs Comparison

The Optimized model requires 43 epochs to achieve its final performance, significantly more than the Baseline (20 epochs) and Lightweight (26 epochs) models. Although the number of epochs is higher, this does not always correlate with improved accuracy, as seen in the Optimized model's results.

3.2. Model Evaluation

The performance of the Convolutional Neural Network (CNN) was evaluated using training and validation metrics for Baseline, resulting in an accuracy of 97%, Optimized 94%, Lightweight 96% (all for test data), followed by a final evaluation on the test dataset. Accuracy and loss values were monitored to evaluate both learning progress and model generalization capabilities.

3.2.1. Training and Validation Accuracy & Loss

3.2.1.1. Baseline Approach

As illustrated in Figure 6, The results of training the CNN model on the baseline configuration show that training and validation accuracy increased consistently to over 97%. However, the loss graph shows that while the training loss continues to decrease steadily, the validation loss tends to fluctuate and begins to increase after the 12th epoch, indicating signs of overfitting. This pattern confirms that while the model is capable of achieving high accuracy on the training data, attention must be paid to the stability of performance on the validation data to ensure good generalization capabilities.

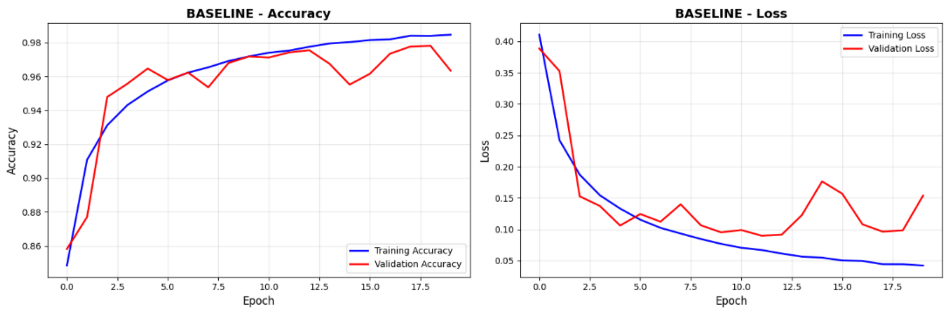

Figure 6. Model evaluation of Baseline Approach (Accuracy & Loss Graph)

3.2.1.2. Optimized Approach

Based on Figure 7, the results of training the CNN model on the optimized configuration show that training accuracy increases gradually and steadily, while validation accuracy tends to be higher but fluctuates at each epoch. In the loss graph, training loss decreases consistently, while validation loss also decreases but still shows significant fluctuations. This pattern indicates that although the model has been optimized and achieves better generalization compared to the baseline, the stability of performance on validation data still needs to be addressed to minimize the potential for overfitting due to variability in validation results.

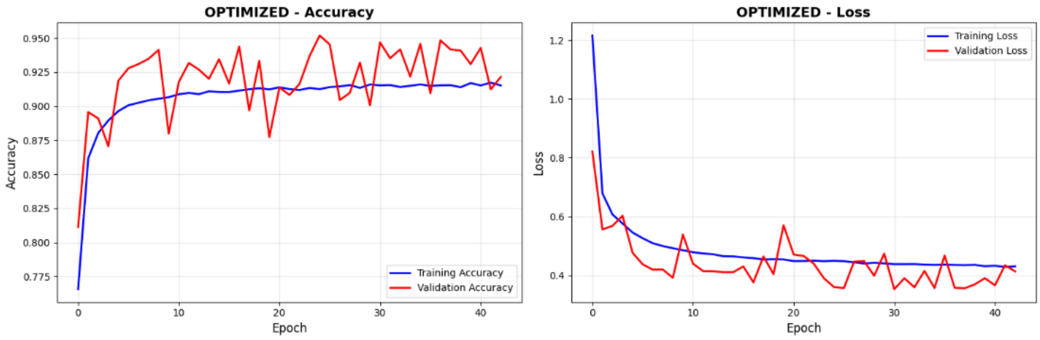

Figure 7. Model evaluation of Optimized Approach (Accuracy & Loss Graph)

3.2.1.3. Lightweight Approach

Based on Figure 8, the results of training the CNN model in the lightweight configuration show that the training accuracy increases gradually and tends to be stable, while the validation accuracy is above the training accuracy but experiences fluctuations at each epoch. In the loss graph, training loss decreases consistently, while validation loss also decreases but with greater variation at certain epochs. This pattern indicates that the lightweight model can achieve good generalization with high validation accuracy, but fluctuations in the validation data suggest potential performance instability that needs to be addressed to reduce the risk of overfitting.

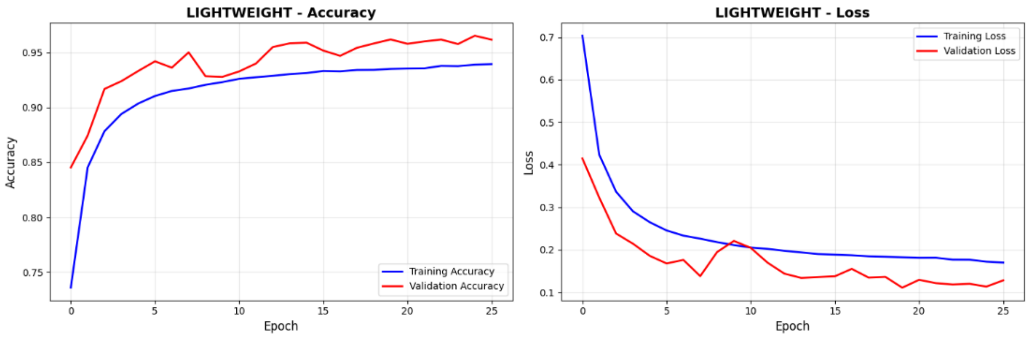

Figure 8. Model evaluation of Lightweight Approach (Accuracy & Loss Graph)

3.3. Confusion Matrix for Testing

The confusion matrix is calculated to further evaluate the classification performance of the proposed CNN-based model for each configuration.

3.3.1. Confusion Matrix for Baseline

As figured in Table 3, The baseline model has a high accuracy rate in all classes, with a main diagonal value close to 10,000. The largest errors occur in class N, which is often predicted as S (243 cases), and class V, which is sometimes predicted as F (203 cases). Overall, the model is quite good, but there are still errors between classes with similar characteristics.

Table 3. Confusion Matrix for Baseline (Testing)

True \ Pred	N	S	V	F	Q
N	9486	243	115	113	43
S	284	9650	60	6	0
V	79	23	9653	203	42
F	0	0	14	9986	0
Q	47	9	12	1	9931

3.3.2. **Confusion Matrix for Optimized** (Testing)

The optimized model has good accuracy per class, but there is a decrease in the number of correct predictions in class N (8852) compared to the previous model. The biggest errors occur in class S, which is predicted as N (632 cases), and class N, which is predicted as S (564 cases). However, the performance in class Q remains very high with minimal errors.

Table 4. Confusion Matrix for Optimized (Testing)

True \ Pred	**N**	**S**	**V**	**F**	**Q**
N	8852	564	103	419	62
S	632	9275	55	38	0
V	114	100	9478	257	51
F	48	23	143	9786	0
Q	27	0	16	8	9949

3.3.3. **Confusion Matrix for Lightweight** (Testing)

The lightweight model also shows high accuracy, but compared to the baseline, the largest errors occur more frequently in class S predicted as N (538 cases) and class N predicted as S (334 cases). Even so, performance in classes F and Q remains stable with few prediction errors.

Table 5. Confusion Matrix for Lightweight (Testing)

True \ Pred	**N**	**S**	**V**	**F**	**Q**
N	9327	334	119	185	35
S	538	9383	46	17	16
V	106	33	9570	263	28
F	18	0	78	9904	0
Q	47	8	39	2	9904

From table 3,4,5 shows that classes F and Q show the highest number of correct predictions across all models, reaching 9,986 (class F, Baseline model) and 9,949 (class Q, Optimized model) respectively, with a very small number of errors. The largest classification error occurred between classes N and S, where class N was most frequently predicted as S 243 times in the Baseline model, 334 times in the Lightweight model, and 564 times in the Optimized model. Conversely, class S was frequently predicted as N, occurring 284 times in the Baseline model, 538 times in the Lightweight model, and 632 times in the Optimized model. Although the Optimized model maintained competitive performance, the number of correct predictions for class N decreased to 8,852, compared to the Baseline model (9,486) and Lightweight model (9,327).

3.4. ROC and AUC

To comprehensively assess the discriminative capability of the Convolutional Neural Network (CNN) classifier, Receiver Operating Characteristic (ROC) curves and their corresponding Area Under the Curve (AUC) values were computed for each arrhythmia class. Given the multi-class nature of the classification task, the One-vs-Rest (OvR) strategy was adopted. In this approach, each class is individually treated as the positive class while the remaining classes are grouped as negative, enabling the derivation of class-specific ROC curves. This methodology has been validated in recent studies as an effective technique for multi-class performance evaluation[17].

The resulting ROC curves are shown in Figure 9,where each line corresponds to a specific class. The ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR), providing a visual interpretation of the classifier's ability to distinguish each class from the others.

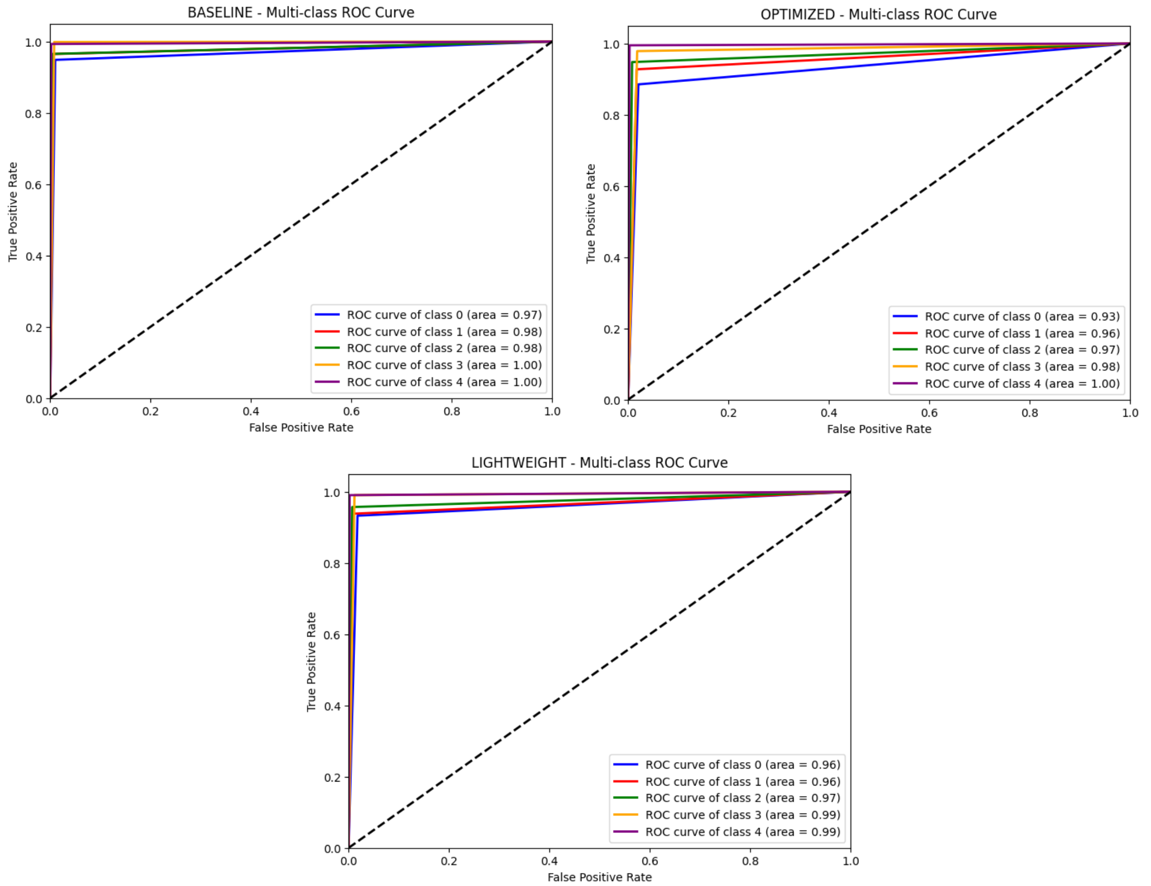

Figure 9. Multi-class ROC Curve showing per-class AUC for Baseline, Optimized, and Lightweight.

As showed in figure 9, Here are some explanations about the figure:

3.4.1. Baseline Model

In the first experiment (Baseline), the CNN model produced an AUC score of 0.97 for class 0, 0.98 for class 1, 0.98 for class 2, 1.00 for class 3, and 1.00 for class 4. The very high AUC values for classes 3 and 4 indicate the model's excellent ability to distinguish between positive and negative data for both classes. Overall, the model's performance across all classes is at a very high level, indicating high sensitivity and specificity across all categories.

3.4.2. Optimized Model

The AUC scores obtained were 0.93 for class 0, 0.96 for class 1, 0.97 for class 2, 0.98 for class 3, and 1.00 for class 4. Compared to the two previous models, there was a slight decrease in class 0, but class 4 still obtained a perfect score. The high performance across all classes indicates that the optimized model still has excellent discriminative ability, despite a slight trade-off in certain classes.

3.4.3. Lightweight Model

The model generated an AUC score of 0.96 for class 0, 0.96 for class 1, 0.97 for class 2, 0.99 for class 3, and 0.99 for class 4. Compared to the Baseline model, the performance was relatively consistent, although there was a slight decrease in some classes. However, the consistently high AUC scores across all classes demonstrate that the lighter architecture still maintains excellent multi-class classification performance.

The Area Under the Curve (AUC) serves as a summary statistic that reflects the overall performance of the classifier. Particularly in multi-class and imbalanced classification problems, AUC offers a robust, threshold-independent evaluation metric[18].

The application of ROC and AUC metrics offers a more detailed evaluation of the classifier's performance beyond simple accuracy measures, especially in multi-class scenarios where the performance across different classes may vary considerably[17]. Moreover, the consistently high AUC values demonstrate the proficiency of CNN architectures in learning complex, hierarchical feature representations that effectively differentiate between closely related classes. This capability aligns with observations reported in recent deep learning classification studies[19].

3.5. Classification Report

The CNN model was evaluated on a configuration using a test dataset consisting of 50,000 ECG samples from five heartbeat classes. Evaluation metrics per class and aggregate, as summarized in Table 4, provide insight into the model's discriminatory ability in all three scenarios. The Baseline achieved an overall accuracy of 97%, the Optimized achieved 94% accuracy, and the Lightweight achieved 96%. This indicates that the Baseline model has the best classification performance, followed by the Lightweight, while the Optimized has the lowest accuracy among the three.

3.5.1. Classification Report for Baseline Approach

Table 6. CNN Classification Metrics per Class for Baseline Approach.

Class	Precision	Recall	F1-Score	Support
N	0.96	0.95	0.95	10000
S	0.97	0.96	0.97	10000
V	0.98	0.97	0.97	10000
F	0.97	1.00	0.98	10000
Q	0.99	0.99	0.99	10000

The overall evaluation metrics for the proposed model are summarized as follows:

a. Accuracy: 97%

b. Macro Average:

   1. Precision: 0.97

   2. Recall: 0.97

3. F1-Score: 0.97

c. Weighted Average:

- Precision: 0.97
- Recall: 0.97
- F1-Score: 0.97

From Table 4, The Baseline Model consistently shows the highest performance across all classes, with an average precision, recall, and F1-score of 0.97. Overall accuracy reaches 97%, with the best performance in classes Q (precision and recall 0.99) and F (recall 1.00), demonstrating excellent detection capabilities for both classes.

3.5.2. Classification Report for Optimized Approach

Table 7. CNN Classification Metrics per Class for Optimized Approach.

Class	Precision	Recall	F1-Score	Support
N	0.92	0.89	0.90	10000
S	0.93	0.93	0.93	10000
V	0.97	0.95	0.96	10000
F	0.93	0.98	0.95	10000
Q	0.99	0.99	0.99	10000

The overall evaluation metrics for the proposed model are summarized as follows:

a. Accuracy: 95%

b. Macro Average:

1. Precision: 0.95
2. Recall: 0.95
3. F1-Score: 0.95

c. Weighted Average:

- Precision: 0.95
- Recall: 0.95
- F1-Score: 0.95

The Optimized model has an overall accuracy of 95% with an average precision, recall, and F1-score of 0.95. Although its performance is still high, there is a decline in class N (recall 0.89), which is the weak point of this model. However, class Q still has the highest score (precision and recall 0.99), indicating excellent classification capabilities in that category.

3.5.3. Classification Report for Lightweight Approach

Table 8. CNN Classification Metrics per Class for Lightweight Approach.

Class	Precision	Recall	F1-Score	Support
N	0.93	0.93	0.93	10000
S	0.96	0.94	0.95	10000
V	0.97	0.96	0.96	10000
F	0.95	0.99	0.97	10000
Q	0.99	0.99	0.99	10000

The overall evaluation metrics for the proposed model are summarized as follows:

a. Accuracy: 965%

b. Macro Average:

1. Precision: 0.96
2. Recall: 0.96
3. F1-Score: 0.96

c. Weighted Average:

- Precision: 0.96
- Recall: 0.96
- F1-Score: 0.96

The Lightweight model achieved an overall accuracy of 96% with an average precision, recall, and F1-score of 0.96. The highest performance was in class Q (precision and recall 0.99) and F (recall 0.99), while class N had a slightly lower score (0.93) but still showed stable results across all metrics.

Table 9. Performance Comparison of Baseline, Optimized, and Lightweight Models

Metric	Baseline	Optimized	Lightweight
Accuracy	0.97	0.95	0.96
Macro Avg Precision	0.97	0.95	0.96
Macro Avg Recall	0.97	0.95	0.96
Macro Avg F1-Score	0.97	0.95	0.96

The evaluation results show that the Baseline model achieves the highest performance with an accuracy of 97%, followed by the Lightweight model at 96%, and the Optimized model at 95%. The Baseline model consistently maintains superior precision, recall, and F1-score across all classes, indicating its strong discriminative ability. Meanwhile, the Lightweight model provides a competitive performance close to the Baseline while potentially offering lower computational costs, making it suitable for resource-constrained environments. The Optimized model, although slightly lower in accuracy, still delivers reliable results and may have advantages in terms of speed or efficiency.

4. Conclusion

Using the MIT-BIH Arrhythmia Database, this study successfully implemented a Convolutional Neural Network (CNN) architecture for ECG-based arrhythmia classification, incorporating extensive preprocessing, data balancing, and augmentation techniques (Time Warping and Magnitude Warping) to improve classification performance across five heartbeat categories. Among the three tested configurations, the Baseline model achieved the highest accuracy of 97% with AUC scores ≥ 0.97 in all classes, demonstrating strong and consistent performance. The Lightweight model had 96% accuracy and better computational efficiency, and the Optimized model had 95% accuracy but reduced performance in the Normal class. These findings demonstrate that the suggested CNN technique is successful in tackling important problems such as feature extraction, class imbalance, and ECG signal fluctuation, while retaining high sensitivity and specificity for arrhythmia identification. The model's persistent high ROC-AUC values across all configurations indicate its excellent discriminative skills, making it promise for speedy, accurate, and reliable automated diagnosis in clinical practice. However, this study has significant limitations, particularly in terms of data balancing; while it significantly improved recognition of minority classes, it also runs the risk of producing a dataset that no longer fully reflects the natural distribution of ECG signals in real-world conditions, which may result in an overestimation of the model's performance and reduced generalization when applied to naturally imbalanced clinical data. Furthermore, the evaluation was restricted to the MIT-BIH dataset, leaving the model's suitability for other datasets or real-time ECG signals unproven. This study differs from previous studies in that it combines Time Warping and Magnitude Warping signal augmentation techniques with directed data balancing, evaluates CNN models in three different configurations using multi-aspect metrics (accuracy, F1-score, ROC-AUC per class), and conducts in-depth classification error analysis. Future study should include testing on many datasets, using adaptive balancing approaches that retain the authenticity of class distributions, and including temporal sequence modeling, such as LSTM or GRU layers, to improve resilience and stability, and real-world applicability.

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